ENHANCED EMERGENCY RESPONSE SYSTEM

19I503-INTERNET OF THINGS

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CONTENTS

COMPONENTS REQUIRED

1.	ESP8266 (NodeMCU)
2.	MQ2 GAS SENSOR
3.	DHT11 SENSOR (Temperature&Humidity)
4.	IR SENSOR
5.	BUZZER
6.	CONNECTING WIRES & BREADBOARD

INTRODUCTION

•**IoT-based Emergency System**: Monitors temperature, humidity, gas, and smoke levels to detect emergencies.

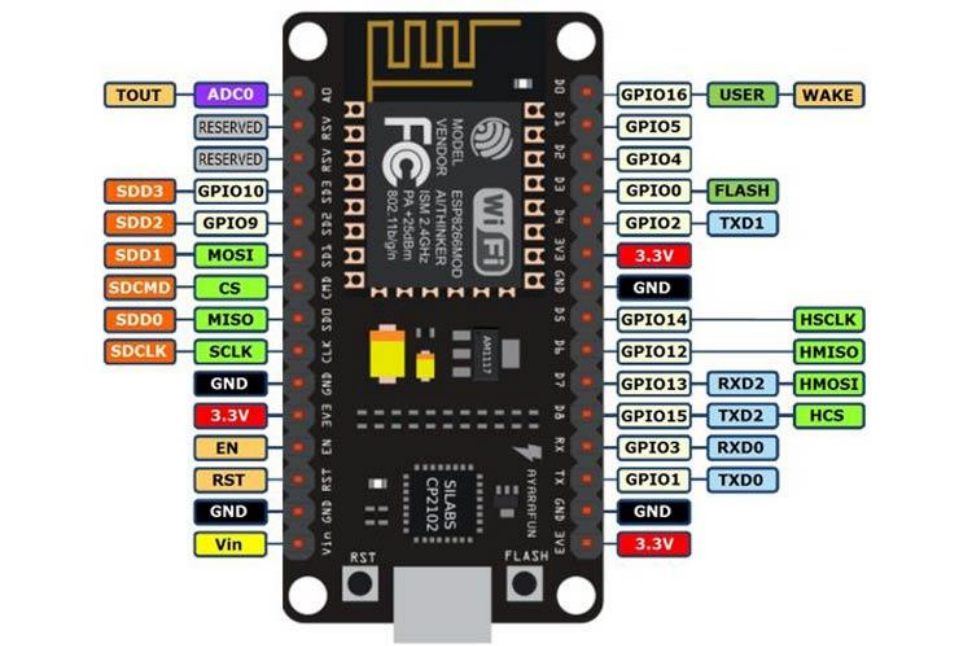
•**Multi-sensor Setup**: Uses DHT11 for temperature/humidity, MQ-2 for gas/smoke, and an IR sensor for motion detection.

•**Real-time Alerts**: Triggers alarms with LED and buzzer when dangerous conditions are detected.

•**Cloud Integration**: Sends sensor data to ThingSpeak for remote monitoring and analysis.

▪**Web application:** integration allows real-time monitoring of sensor data, providing users with instant access to environmental conditions.

ARCHITECTURE OF ESP8266

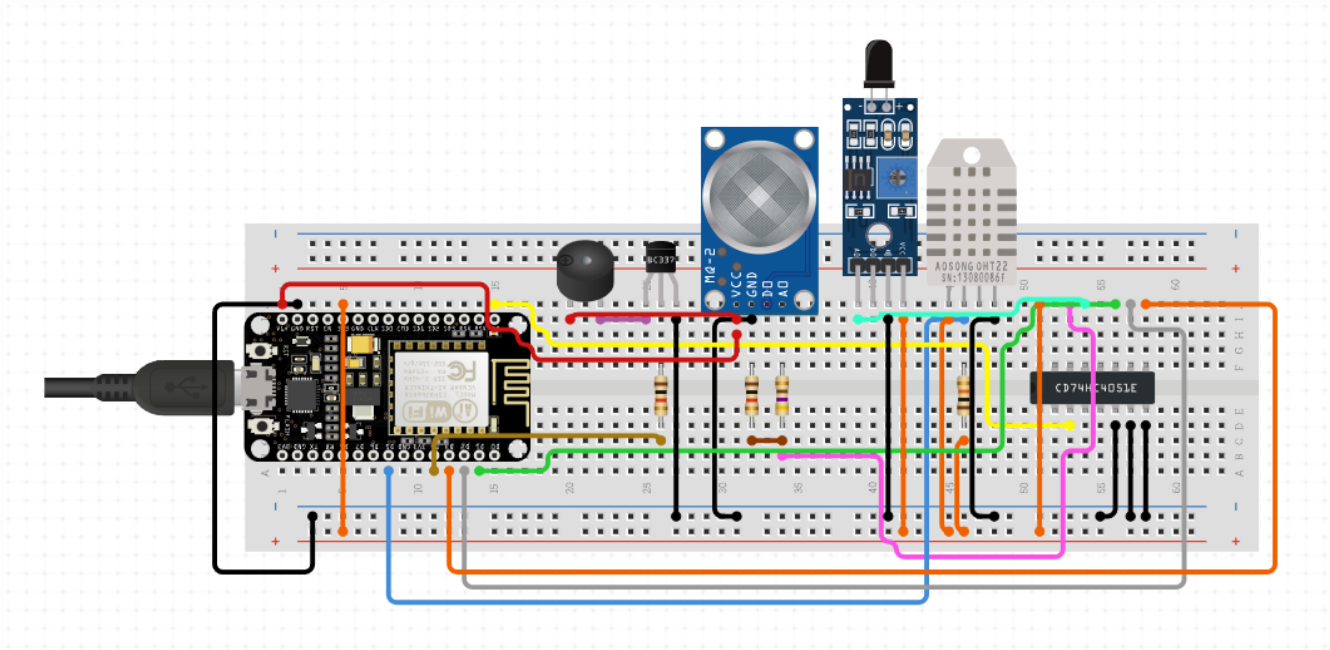


ARCHITECTURE OF ESP8266

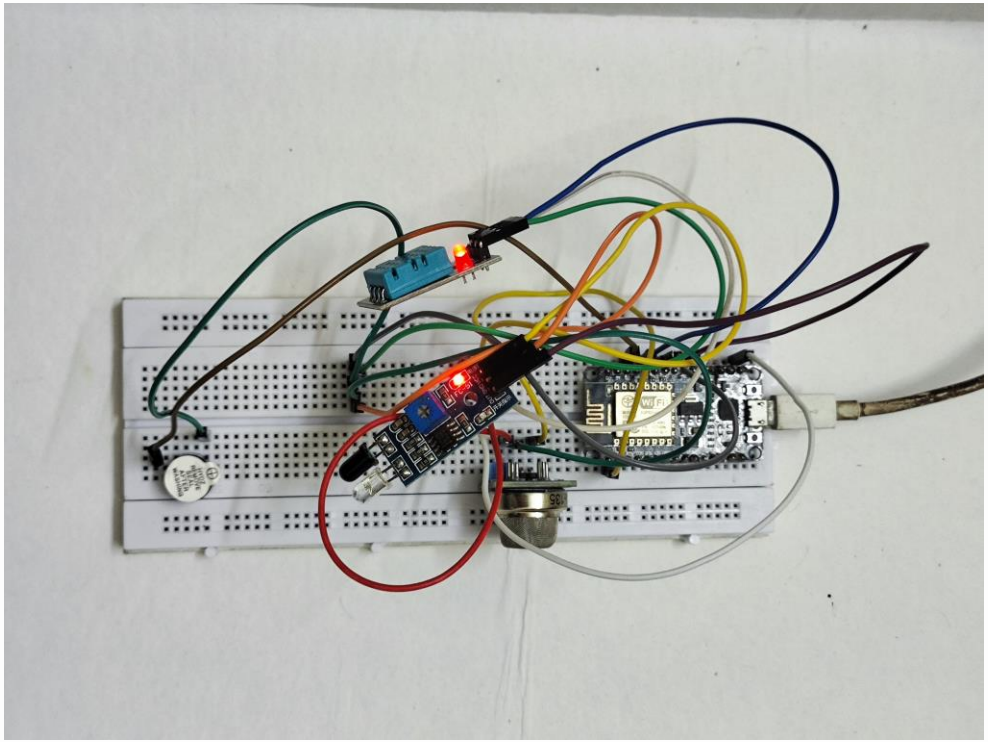
The ESP8266 microcontroller is designed for efficient wireless communication and processing:

- **CPU**: A low-power 32-bit processor (80-160 MHz) that handles Wi-Fi, GPIO control, and user applications.
- **Wi-Fi Module**: Supports 2.4 GHz Wi-Fi for network connectivity, with an integrated TCP/IP stack for web communication.
- **Memory**: 160 KB RAM for processing, and up to 4 MB flash storage for firmware and code.
- **GPIO**: 16 pins for connecting sensors and devices, supporting I2C, SPI, UART, and PWM.
- **Power Management**: Low-power modes for battery use, operates at 3.3V.
- **ADC**: One analog input pin for sensors.

CONNECTIONS



CONNECTIONS



WORKING PRINCIPLES

•**Sensor Monitoring:** The system continuously monitors environmental data using sensors like DHT11 for temperature and humidity, MQ-2 for smoke/gas, and an IR sensor for object detection.

•**Threshold Detection:** The system compares real-time sensor values against predefined threshold values (e.g., temperature above 30°C, high gas levels). If any values exceed these limits, an alert is triggered.

•**Buzzer and LED Activation:** Upon detecting hazardous conditions (such as smoke or high temperature), the buzzer and LED are activated to provide immediate local warnings.

•**Data Upload to ThingSpeak:** Sensor data is transmitted to the ThingSpeak IoT platform, where it can be stored, analyzed, and visualized in real-time.

•**Web App Integration:** A web application is developed to link with ThingSpeak, allowing users to view live readings and trends from the sensors, enhancing remote monitoring capabilities.

•**Real-Time Remote Monitoring:** The uploaded data and web app interface enable users to remotely monitor environmental conditions and respond quickly during emergencies.

ARDUINO SKETCH

```
#include <ESP8266WiFi.h>
#include <WiFiClient.h>
#include <ESP8266HTTPClient.h>
#include <DHT.h>

// Define pins for sensors and outputs
#define DHTPIN D4              // Pin where the DHT11 is connected
#define DHTTYPE DHT11          // DHT11 sensor
#define BUZZER_PIN D2          // Pin for the buzzer
#define LED_PIN D5             // Pin for the LED
#define SMOKE_SENSOR_PIN A0    // Analog pin for the smoke sensor (MQ-2)
#define GAS_SENSOR_PIN A0      // Analog pin for the gas sensor (MQ-135) - same as smoke sensor for testing
#define IRSENSORPIN D6         // IR sensor pin

// Initialize the DHT11 sensor
DHT dht(DHTPIN, DHTTYPE);

// ThingSpeak API credentials
const char* writeAPIKey = "DPPM6BOSWIR21G86";  // Replace with your ThingSpeak API key
const char* server = "api.thingspeak.com";

// WiFi credentials
const char* ssid = "12341234";          // Your WiFi SSID
const char* password = "123412345";  // Your WiFi Password

// Set threshold values
const float TEMP_THRESHOLD = 30.0;  // Temperature threshold in °C
const float HUMIDITY_THRESHOLD = 70.0; // Humidity threshold in %
const int MQ2_THRESHOLD = 300;       // MQ-2 gas/smoke threshold (adjust based on calibration)

WiFiClient client;
```

ARDUINO SKETCH

```
void setup() {
  // Initialize serial communication
  Serial.begin(115200);

  // Initialize sensor and output pins
  pinMode(BUZZER_PIN, OUTPUT);
  pinMode(LED_PIN, OUTPUT);
  pinMode(IRSENSORPIN, INPUT);  // Set IR sensor pin as input

  // Start DHT sensor
  dht.begin();

  // Connect to WiFi
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
    Serial.println("Connecting to WiFi...");
  }
  Serial.println("Connected to WiFi");
}

void loop() {
  // Read temperature and humidity from DHT11 sensor
  float temperature = dht.readTemperature();
  float humidity = dht.readHumidity();

  // Read values from smoke sensor (MQ-2)
  int smokeLevel = analogRead(SMOKE_SENSOR_PIN);

  // Read IR sensor for object detection
  int irValue = digitalRead(IRSENSORPIN);
  int objectDetected = (irValue == LOW) ? 1 : 0;  // 1 for detected, 0 for not detected
```

ARDUINO SKETCH

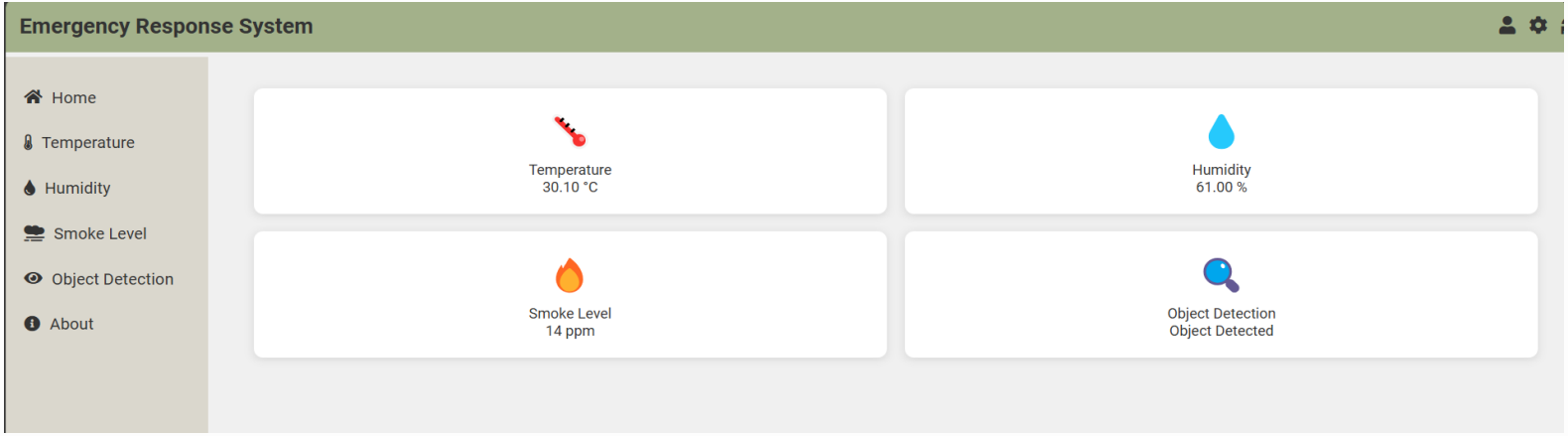
```
  // Print sensor data to the serial monitor
  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.println(" °C");

  Serial.print("Humidity: ");
  Serial.print(humidity);
  Serial.println(" %");

  Serial.print("Smoke Level: ");
  Serial.println(smokeLevel);

  Serial.print("Object Detection: ");
  Serial.println(objectDetected ? "Object detected" : "No object detected");

  // Check thresholds and take action
  if (temperature > TEMP_THRESHOLD || humidity > HUMIDITY_THRESHOLD || smokeLevel > MQ2_THRESHOLD) {
    Serial.println("WARNING: Threshold exceeded!");
    digitalWrite(BUZZER_PIN, HIGH);
    digitalWrite(LED_PIN, HIGH);

    // Send data to ThingSpeak
    sendToThingSpeak(temperature, humidity, smokeLevel, objectDetected);
  } else {
    digitalWrite(BUZZER_PIN, LOW);
    digitalWrite(LED_PIN, LOW);
  }

  // Wait for 2 seconds before the next reading
  delay(2000);
}

void sendToThingSpeak(float temperature, float humidity, int smokeLevel, int objectDetected) {
  if (WiFi.status() == WL_CONNECTED) {
    HTTPClient http;
```

ARDUINO SKETCH

```
    // Build URL for the GET request
    String url = "http://" + String(server) + "/update?api_key=" + String(writeAPIKey) +
                 "&field1=" + String(temperature) +
                 "&field2=" + String(humidity) +
                 "&field3=" + String(smokeLevel) +
                 "&field4=" + String(objectDetected);  // New field for object detection

    // Print URL to serial for debugging
    Serial.print("Sending data to URL: ");
    Serial.println(url);

    // Send the request to ThingSpeak
    http.begin(client, url);
    http.setTimeout(10000);  // Set timeout to 10 seconds
    int httpResponseCode = http.GET();

    // Check for HTTP errors
    if (httpResponseCode > 0) {
      Serial.print("ThingSpeak Response code: ");
      Serial.println(httpResponseCode);
      if (httpResponseCode == 200) {
        Serial.println("Data sent successfully to ThingSpeak!");
      } else {
        Serial.println("Error: Non-200 response from ThingSpeak.");
      }
    } else {
      Serial.print("Error sending data to ThingSpeak: ");
      Serial.println(http.errorToString(httpResponseCode));
    }
    http.end(); // Free resources
  } else {
    Serial.println("WiFi not connected.");
  }
}
```

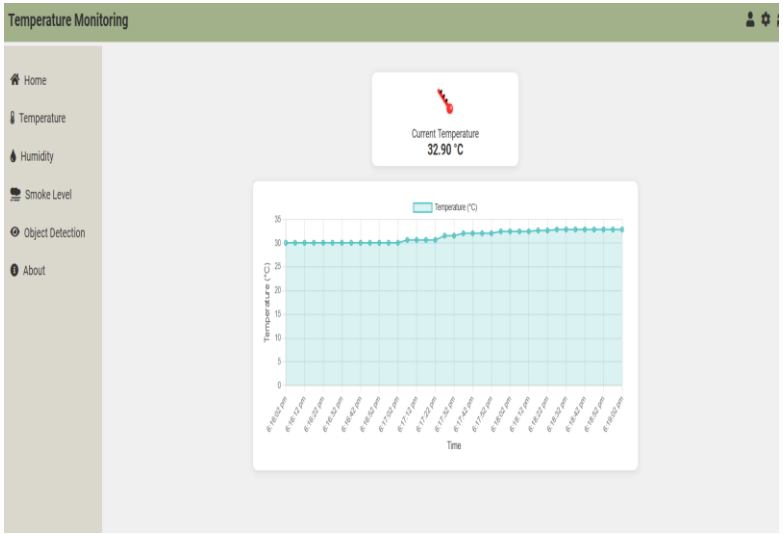
WEB APPLICATION

DASHBOARD

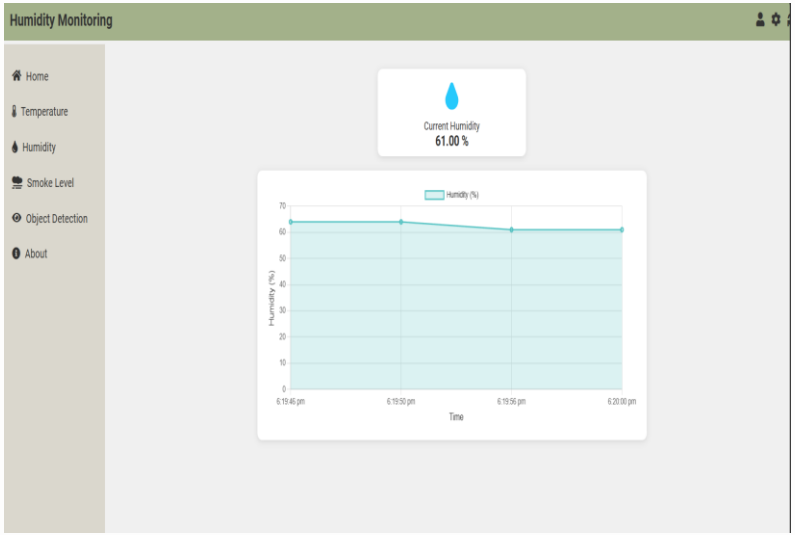


WEB APPLICATION

TEMPERATURE



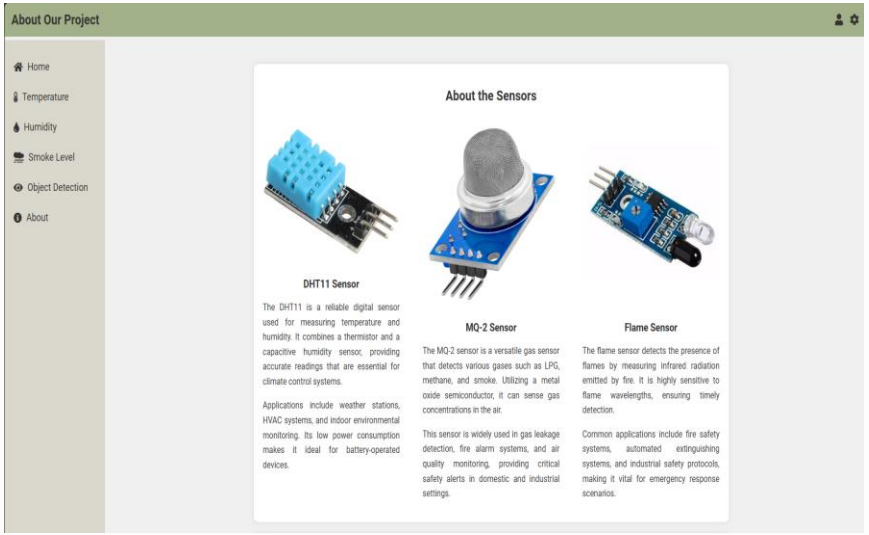
HUMIDITY



WEB APPLICATION

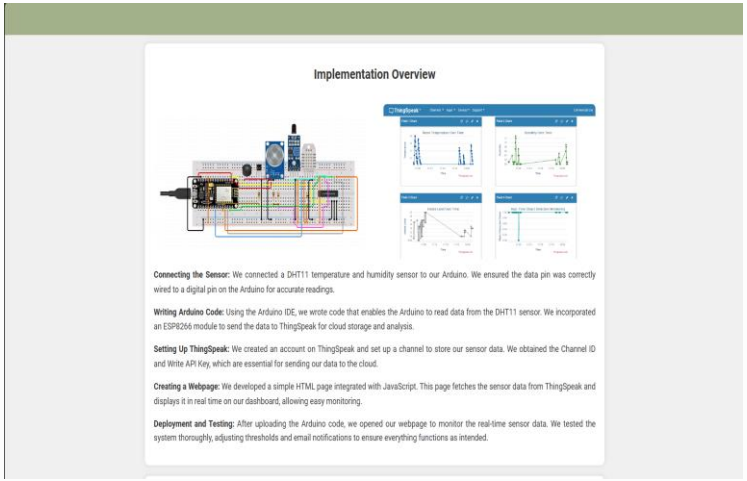
SMOKE LEVEL

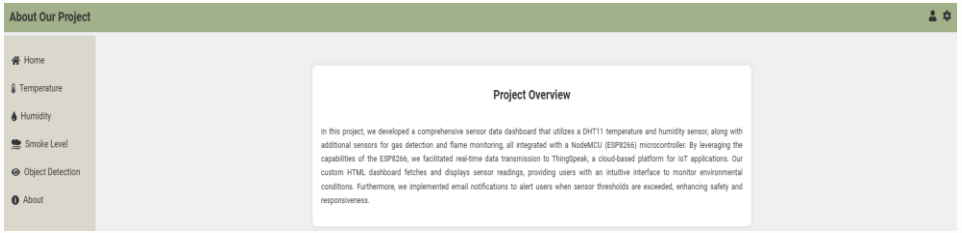
COMPONENTS USED



WEB APPLICATION

ABOUT





APPLICATIONS

•**Smart Home Security**: Monitors smoke and gas levels for enhanced safety and immediate alerts to homeowners and emergency services.

•**Industrial Safety Monitoring**: Detects hazardous conditions in factories to ensure worker safety and regulatory compliance.

•**Environmental Monitoring**: Tracks soil moisture and air quality for better agricultural management and weather risk reduction.

•**Disaster Management**: Provides real-time data during natural disasters to help emergency responders allocate resources effectively.

•**Healthcare Facilities**: Ensures optimal temperature and humidity in hospitals to prevent pathogen growth, particularly in sensitive areas.

•**Public Buildings and Infrastructure**: Detects emergencies like fires or gas leaks promptly to safeguard large groups in schools, malls, and transport systems.

CONCLUSION

The Enhanced Emergency Response System effectively monitors environmental conditions using DHT11, MQ-2, and IR sensors to detect hazards such as high temperatures and gas leaks. By employing threshold detection, the system triggers local alerts via buzzer and LED activation when predefined limits are exceeded.

Data is transmitted to the ThingSpeak IoT platform for real-time monitoring and visualization, and also provides useful insights about the situation with the help of the web application allowing users to respond quickly to emergencies. This project showcases the practical application of IoT technology in enhancing safety and emergency preparedness.

THANK YOU